Question			Marking details	Marks available					
				AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
7	*(a)*	(i)	Work done = 65 000 × 32 = 2.08×10^6 J		1		1	1	
		(ii)	Energy at B = 2 600 × 9.81 × 42 (1) or 1.07×10^6 seen Substitution into % efficiency = $\frac{1.07 \times 10^6 \times 100\%}{2.08 \times 10^6}$ (1) (**no ecf**) % efficiency = 51.5% (1) [accept 51% – 55%, or 0.51 – 0.55]	1	1 1		3	3	
	(b)		Work against resistive forces = 2 800 × 36 (1) or 101 kJ seen ΔE_p using 30 m drop = -765 kJ (1) Substitution into work-energy theorem: KE gain = 765 kJ – 101 kJ = 664 kJ (1) 664 000 = ½ × 2 600 × v^2 (1) [allow use of 765 kJ] v = 22.6 m s^{-1} (1) [765 kJ → 24.3 m s^{-1}] Additional marking guidance • Whole drop (42 m) with resistive forces → 27.3 m s^{-1} → 4 marks • 30 m drop without resistive forces → 24.3 m s^{-1} → 3 marks • Whole drop without resistive forces → 28.7 m s^{-1} → 2 marks • Mixing force and energy → 0 • Use of $v^2 = u^2 + 2ax$ →0 • Dissipate energy =	1	1 1 1 1		5	5	
			Question 7 total	**2**	**7**	**0**	**9**	**9**	**0**

© WJEC CBAC Ltd.